USN | | | | | | | | | | [23CSE102]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY -2024

OBJECT-ORIENTED PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	List out the types of operators used in Java.	L1	CO1	**02**
	b.	Define an object and Class.	L2	CO2	**02**
	c.	Discuss the usage of aggregation.	L3	CO3	**02**
	d.	What are rules to create method overloading?	L2	CO4	**02**
	e.	Define an Exception.	L1	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	What is Java? Explain the features of Java	L1	CO1	**05**
		OR			
	b.	List out the looping statements available in Java. Explain with an example.	L2	CO1	**05**
3.	**a.**	Explain the four types of access modifiers and its access levels with suitable examples.	L2	CO2	**05**
		OR			
	b.	Explain the different types of packages used in Java with some examples and how we can access one package from another package.	L2	CO2	**05**
4.	**a.**	Write down the usage of super Keyword with an example.	L2	CO3	**05**
		OR			
	b.	Define an Aggregation. Explain with an example program.	L2	CO3	**05**
5.	**a.**	Differentiate method overriding and method overloading.	L2	CO4	**05**
		OR			
	b.	Compare tight coupling and loose coupling.	L2	CO4	**05**
6.	**a.**	Explain in detail about throw and throws keyword used in Exception handling.	L3	CO5	**05**
		OR			
	b.	Differentiate between checked and unchecked exception.	L2	CO5	**05**
7.	**a.**	Explain the concepts of arrays in Java with the steps of array declaration and creation. How can one dimensional array can be initialized.	L2	CO1	**05**
		OR			
	b.	Explain the different types of integer data types.	L2	CO1	**05**

		PART-C	**06 X 10=60**		
8.	**a.**	Explain in detail about basic Concepts of OOPs.	L2	CO1	**10**
		OR			
	b.	Explain switch statement with syntax and example. Develop a java program that prompts the user to enter two floating point (double) numbers and an operator(*, +, /, %, -). Print the results of the given operation.	L3	CO1	**10**
9.	**a.**	What is Constructor Chaining? How do we achieve Constructor Chaining? Explain with an example.	L3	CO2	**10**
		OR			
	b.	Write a java program to i) To demonstrate the usage of parameterized constructor. ii) To find the area of rectangle using the concept of classes and objects	L3	CO2	**10**
10.	**a.**	Explain in detail about Multilevel Inheritance with an example program.	L2	CO3	**10**
		OR			
	b.	What is an interface? Explain the rules to create an interface in java with example.	L2	CO3	**10**
11.	**a.**	Explain in detail about compile time polymorphism with suitable program.	L2	CO4	**10**
		OR			
	b.	What is runtime polymorphism? How to achieve this? Explain with program.	L3	CO4	**10**
12.	**a.**	Differentiate error and exception. Develop a java program which contains a method that throws Divide by Zero Exception and use suitable exception handler so that exception can be displayed	L3	CO5	**10**
		OR			
	b.	Explain multiple catch clauses with its general syntax and the usage of finally block with an example.	L3	CO5	**10**
13.	**a.**	Explain in detail about uses of getters and setters method.	L2	CO2	**10**
		OR			
	b.	Explain the different types of inheritance and give the importance of the keyword extends in inheritance. Write a java Program to implement Banking application using **single inheritance**. Create a super class, get the acc no, name, balance. In sub class get the amount to be withdrawn. Display the balance amount after withdraw.	L3	CO2	**10**